

Data Science

Lecture 02: Probability Theory

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Module 0

Orientation

Guiding Question: *What does a data scientist actually need from probability theory — and why does the formal treatment come first?*

The Landscape

Probability theory

Given a model of chance,
what follows?

Kolmogorov axioms,
distributions, limit theorems

Mathematical statistics

Given data, which model of
chance is plausible?

Estimation, hypothesis testing

Data science

Given data and a decision,
what should we do?

Prediction, learning,
evaluation

The arrow runs left to right: this module builds the vocabulary the rest of the course spends its time using.

What We Need From Probability

In this module

- sample spaces and events
- counting outcomes
- probability as a measure
- conditioning and independence
- Bayes' theorem
- random variables and distributions
- describing data numerically
- testing a hypothesis

Where it comes back

- naive Bayes and Bayesian decisions
- impurity measures in decision trees
- likelihoods in probabilistic models
- sampling and error estimates in evaluation
- independence assumptions in graphical models

Example: Two Fruit Boxes

Red box

2 apples, 6 oranges

Blue box

3 apples, 9 oranges

We pick a box, then a fruit from it.

- If we pick the **red box** **40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

The Counts, in One Grid

Twenty picks, recorded as a table of counts — box across, fruit down:

	$B = r$	$B = b$	row total
$F = a$ (apple)	2	3	5
$F = o$ (orange)	6	9	15
column total	8	12	20

Joint

a single cell over the total

$$P(B = r, F = a) = \frac{2}{20}$$

Marginal

a row or column total over the total

$$P(F = a) = \frac{5}{20}$$

Conditional

a cell over its own row or column

$$P(F = a \mid B = r) = \frac{2}{8}$$

The Two Rules

Sum rule

$$P(X) = \sum_Y P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y | X)P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X \mid Y)P(Y)$$

Which box did the orange come from?

Bayes' Theorem, in Advance

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$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X \mid Y)P(Y)$$

Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{0.75 \cdot 0.4 + 0.75 \cdot 0.6} = 0.4$$

Here the fruit told us **nothing** about the box — both boxes have the same orange fraction. Evidence is only informative when the hypotheses predict different things.

Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. The rest of this chapter makes them precise and extends them beyond counting.

Module 1

Events and the Algebra of Sets

Guiding Question: *If probability is going to be a measure, what exactly are the objects it measures — and how do they combine?*

What Makes an Experiment "Random"?

Definition (Random experiment). A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Randomness as Missing Information

A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.

Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Sample Space and Events

Definition (Sample space, event space). A set $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$. The elements are called **outcomes**.

Let Ω be finite. Every subset $A \subseteq \Omega$ is an **event** the event A **occurs** iff the observed outcome ω is a member of A . The set of all events, the power set $\mathcal{P}(\Omega)$, is the **event space**.

An outcome is a single result; an event is a *question you can ask* about the result.

For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set we return to this when continuous distributions appear.

Important Event Types

$$\emptyset$$

impossible event

$$|A| = 1$$

elementary event

$$\Omega$$

certain event

$$A \subseteq B$$

A is a sub-event of B — " A entails B "

$$\overline{A} := \Omega \setminus A$$

complementary event

$$A \cap B = \emptyset$$

A and B are incompatible

Examples

One coin flip

$$\Omega = \{H, T\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

Event "heads": $E = \{H\}$

Two coin flips

$$\Omega = \{HH, HT, TH, TT\}$$

Event "at least one head": $E = \{HH, HT, TH\}$

Four outcomes, sixteen possible events.

The Event Space Grows Fast

Three-sided die: $\Omega = \{1, 2, 3\}$, so $\mathcal{P}(\Omega)$ has $2^3 = 8$ events, ordered by inclusion:

$\{1, 2, 3\}$ $\{1, 2\}$ $\{1, 3\}$ $\{2, 3\}$ $\{1\}$ $\{2\}$ $\{3\}$ $\emptyset$

For $|\Omega| = n$ the event space has 2^n elements — which is exactly why the next unit is about counting.

Not Every Sample Space Is Finite

Dosage of a drug

$$\Omega = (0, \infty)$$

Uncountably many outcomes.

Race with 7 teams

$$\Omega = \{\text{all orderings of } 1, \dots, 7\}$$

$7! = 5040$ outcomes.

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Venn Diagrams

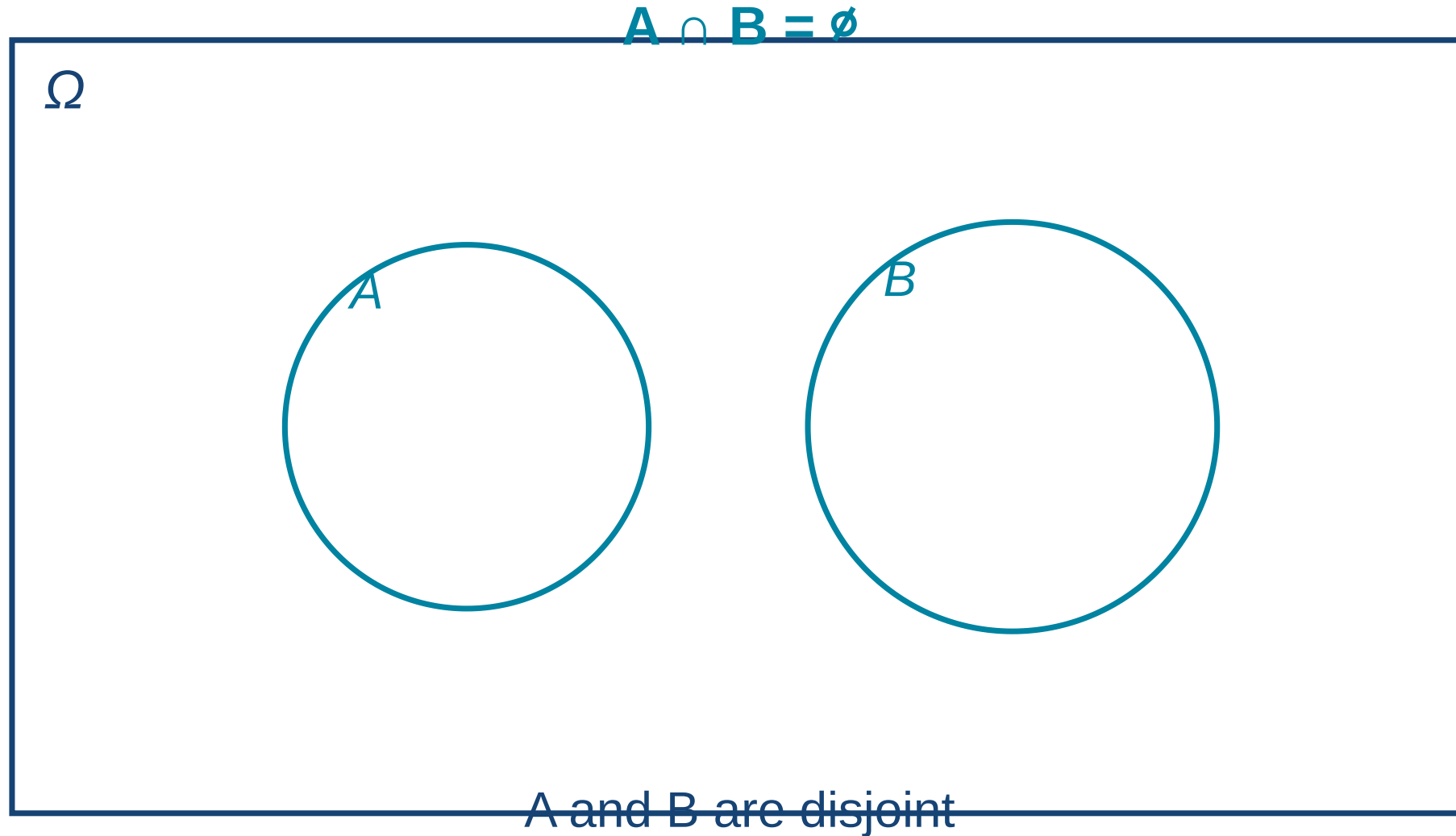
Venn diagrams are the graphical representation of events:

- the rectangle represents the sample space Ω
- circles represent events $A \subseteq \Omega, A \in \mathcal{P}(\Omega)$

Every region of the picture *is* a set — and every set operation is a region you can point at.

Important Event Types

Before each step: which region will light up?

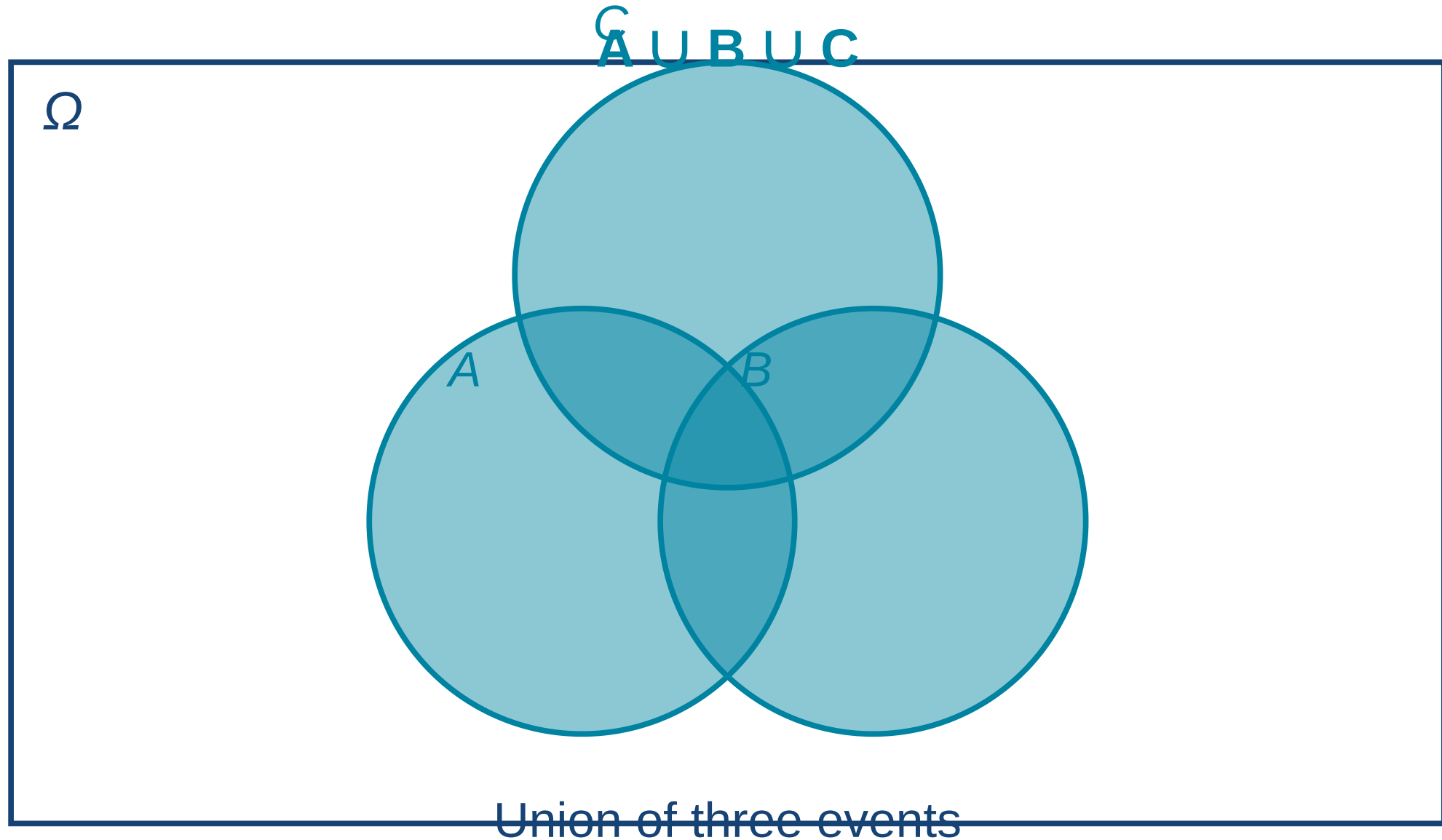


Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions



Mini-Exercise

Sketch (pen and paper) the regions for:

1. $(A \cup B) \cap C$
2. $(A \cap B) \cup C$

Are they the same region? Which algebra law says so — or says they are not?

Mini-Exercise — Solution

Mini-Exercise — Solution

They differ: $(A \cup B) \cap C$ lies entirely **inside** C , while $(A \cap B) \cup C$ contains **all** of C plus the lens $A \cap B$ outside it.

The distributive laws expand each into a different expression — the algebra distinguishes exactly what the picture shows.

Module 2

Counting

Guiding Question: How do we determine the size of a sample space we could never write down?

Why Counting Comes First

If every outcome in a finite sample space is equally likely, then the probability of an event is a **ratio of counts**:

$$P(A) = \frac{|A|}{|\Omega|}$$

So before we can measure probability, we must be able to count $|A|$ and $|\Omega|$ — often without enumerating them.

Basic Principle of Counting

Generalised basic principle of counting. If an experiment consists of r stages, where stage i has n_i possible outcomes regardless of the earlier stages, then the whole experiment has

$$n_1 \cdot n_2 \cdots n_r$$

possible outcomes.

Example. How many 7-place licence plates, if the first 3 places are letters (26) and the last 4 are digits (10)?

$$26^3 \cdot 10^4 = 175,760,000$$

Three Questions Decide the Formula

Order?

Does the sequence matter?
matters → **permutation**
(rankings)
does not → **combination**
(committees)

Repetition?

May an element be picked twice?
without — teams of people
with — letters in a word

How many?

Select k of n elements, or arrange
all n ?

The Four Basic Cases

Ordered, without repetition

k -permutation of n :

$$\frac{n!}{(n-k)!}$$

From $\{a, b\}$, $k = 2$: ab, ba

Ordered, with repetition

$$n^k$$

From $\{a, b\}$, $k = 2$: aa, ab, ba, bb

Unordered, without repetition

k -combination of n :

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

From $\{a, b\}$, $k = 2$: $\{a, b\}$

Unordered, with repetition

$$\binom{n+k-1}{k}$$

From $\{a, b\}$, $k = 2$: $\{a, a\}, \{a, b\}, \{b, b\}$

Permutations

Definition (k -permutation). An ordered selection of k elements from a set of n distinct elements without repetition. Counting position by position: n choices, then $n - 1$, ..., down to $n - k + 1$, giving $\frac{n!}{(n-k)!}$.

Example. A class of 6 men and 4 women is ranked after an exam, no ties. How many possible rankings? $10! = 3,628,800$.

Combinations

Each k -combination corresponds to $k!$ different k -permutations — the same k elements in every possible order. Therefore

$$\binom{n}{k} = \frac{\text{number of } k\text{-permutations}}{k!} = \frac{n!}{k!(n-k)!}$$

Example. A committee of 3 from 20 people: $\binom{20}{3} = 1140$.

Dividing by $k!$ is the standard move for "forgetting" an ordering we counted.

Remember it as *over-count, then correct*.

Properties of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Choosing which k to include is the same as choosing which $n - k$ to leave out.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Split by a fixed element i : groups **containing** i , plus groups **not containing** i .

Binomial theorem.

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Application: How Many Subsets?

A set with n elements has

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

subsets — sum over all subset sizes.

Alternative intuition. Each element is either in the subset or not: a binary choice, n times, so $2 \cdot 2 \cdots 2 = 2^n$.

This is the size of the event space $\mathcal{P}(\Omega)$ from the previous unit.

Mini-Exercise

A password consists of 4 characters from a 26-letter alphabet.

1. How many passwords are there?
2. How many with **no repeated letter**?
3. How many **sets** of 4 distinct letters (order ignored)?

Mini-Exercise — Solution

Mini-Exercise — Solution

1. Ordered, with repetition

$$26^4 = 456,976$$

2. Ordered, no repetition

$$\frac{26!}{22!} = 358,800$$

3. Unordered, no repetition

$$\binom{26}{4} = 14,950$$

Takeaway

Counting reduces to three questions — **order?** **repetition?** **how many?** — and one habit: when an ordering was counted that should not have been, divide it out.

Module 3

What Is a Probability?

Guiding Question: *Every attempt to define probability turns out to be circular. What do we do instead?*

An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of A is how likely the stone lands inside A , assuming

1. we hit the rectangle Ω for certain, and
2. every part of Ω is equally likely to be hit.

Probability is then a **measure of the state** $A \subseteq \Omega$ of a random experiment — literally, how much of the sample space the event occupies.

Both assumptions are doing real work. Assumption 2 is the one we will have to give up.

Two Ways to Read a Probability

Frequentist

The probability of an elementary event a is the fraction of repetitions of the experiment that yield a .

Requires the experiment to be repeatable.

Bayesian

The probability of a is the degree of certainty that a will happen — even without conducting the experiment.

Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment** its probabilities need not be estimated, they are implied by the setup.

Why That Definition Is Not Enough

The circularity. "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \rightarrow \mathbb{R}$ is a **probability measure** if:

Axiom I

$$P(A) \geq 0$$

non-negative

Axiom II

$$P(\Omega) = 1$$

normalised

Axiom III

$$A \cap B = \emptyset \Rightarrow$$

$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

Beyond finite sample spaces. When Ω is infinite, the event space is not the full power set but a σ -**algebra** of measurable sets, and Axiom III is required for **countably many** disjoint

Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

(Axioms II, III)

$$P(\emptyset) = 0$$

(from the above with $A = \Omega$)

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity *(Axioms I, II)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion *(Axiom III)*

For mutually exclusive $A_1, \dots, A_p$: $P(A_1 \cup \dots \cup A_p) = P(A_1) + \dots + P(A_p)$.

Takeaway

Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Laplace Probabilities in Practice

When all outcomes in Ω are equally likely, computing $P(A)$ means **counting twice**: the favourable cases, and the possible ones.

$$P(A) = \frac{|A|}{|\Omega|}$$

The whole difficulty is choosing a sample space in which "equally likely" is actually true.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

Why ordered pairs?
Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

- $|\Omega| = 6 \cdot 6 = 36$ ordered pairs
- $E = \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$, so $|E| = 6$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Why ordered pairs?
Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Ordered or Unordered — Both Work

An urn holds **6 white** and **5 black** balls. Three are drawn. **What is the probability of exactly 1 white and 2 black?**

Variant 1 — order matters

$$|\Omega| = 11 \cdot 10 \cdot 9 = 990$$

Favourable: white-black-black, black-white-black, black-black-white = $6 \cdot 5 \cdot 4 + 5 \cdot 6 \cdot 4 + 5 \cdot 4 \cdot 6 = 360$

$$\frac{360}{990} = \frac{4}{11}$$

Variant 2 — order ignored

$$|\Omega| = \binom{11}{3} = 165$$

Favourable: $\underbrace{\binom{6}{1}}_{\text{the white}} \cdot \underbrace{\binom{5}{2}}_{\text{the blacks}} = 6 \cdot 10 = 60$

$$\frac{60}{165} = \frac{4}{11}$$

The General Rule

When selecting k items from n , you may count **with** or **without** order — as long as you are consistent, and as long as the chosen sample space really is equiprobable.

Pick whichever counting is easier for the event at hand.

Mini-Exercise

Five people are chosen from **10 couples** (20 individuals).

What is the probability that the five chosen are **mutually unrelated** — no couple is fully represented?

Mini-Exercise — Solution

Mini-Exercise — Solution

Choose 5 of the 10 couples, then one person from each:

$$P(N) = \frac{\binom{10}{5} \cdot 2^5}{\binom{20}{5}} = \frac{252 \cdot 32}{15504} \approx 0.52$$

Order is ignored consistently in numerator and denominator.

Module 4

Conditioning, Independence, Bayes

***Guiding Question:** A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what is the correct bookkeeping?*

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll: $\Omega = \{(1, 1), (1, 2), \dots, (6, 6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3, 1), \dots, (3, 6)\}$ remains
- Outcomes like $(2, 4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3, 5)\}$

$$P = \frac{1}{6}$$

Definition

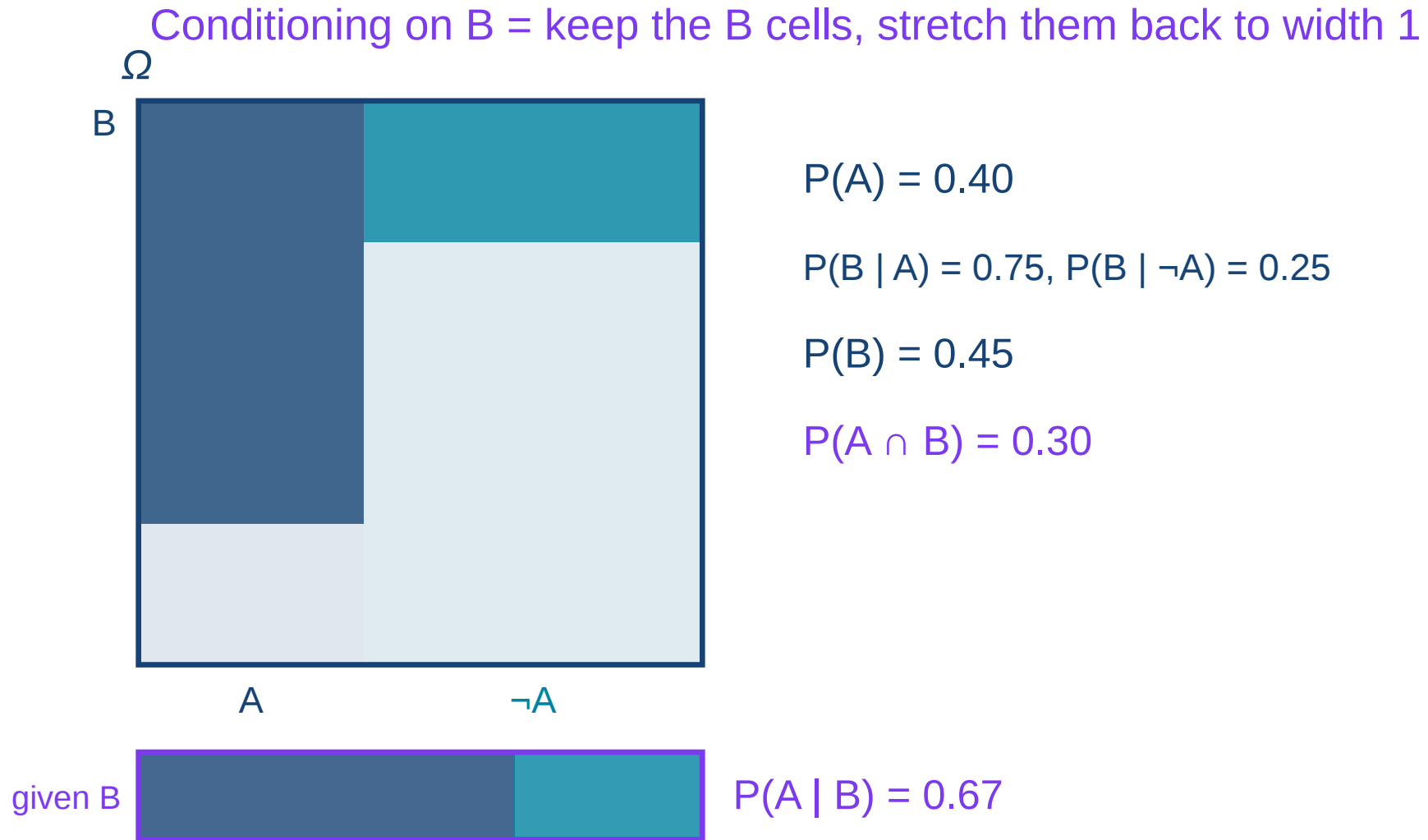
Definition (Conditional probability).

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) > 0$$

$P(A \mid B)$ is the probability of A **under the condition** B .

Back to the stone: *given that the stone landed in area B , how likely is it that it also landed in A ?*
We renormalise by the area of B , because B is now the whole world.

Conditioning, Drawn



Before the last click: what must the two shaded cells be divided by?

Example: Two Coins

Two coins are flipped. The event of interest is $B = \{(h, h)\}$.

(a) "The first coin is heads"

$$F = \{(h, h), (h, t)\}$$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$$A = \{(h, h), (h, t), (t, h)\}$$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event B , two different pieces of information, two different answers. **Why is (b) smaller?**

Total Probability

Theorem (Total probability). Let $B_1, \dots, B_k$ be mutually exclusive with $\Omega = B_1 \cup \dots \cup B_k$ and $P(B_i) > 0$. Then for any event A :

$$P(A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

Proof. Decompose A along the partition and use additivity:

$$P(A) = P((B_1 \cup \dots \cup B_k) \cap A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

Rules That Follow

Product rule

$$\begin{aligned}P(A \cap B) &= P(B) \cdot P(A \mid B) \\ &= P(A) \cdot P(B \mid A)\end{aligned}$$

Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B , the map $A \mapsto P(A \mid B)$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B .

Chain Rule

$$P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \dots \cap E_{n-1})$$

By successive application of the product rule.

Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Takeaway

Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \quad \text{and} \quad P(A \mid \overline{B}) = P(A)$$

Independence means: **learning B changes nothing about A .**

The Classic Confusion

Statistical independence does not mean the events are disjoint.

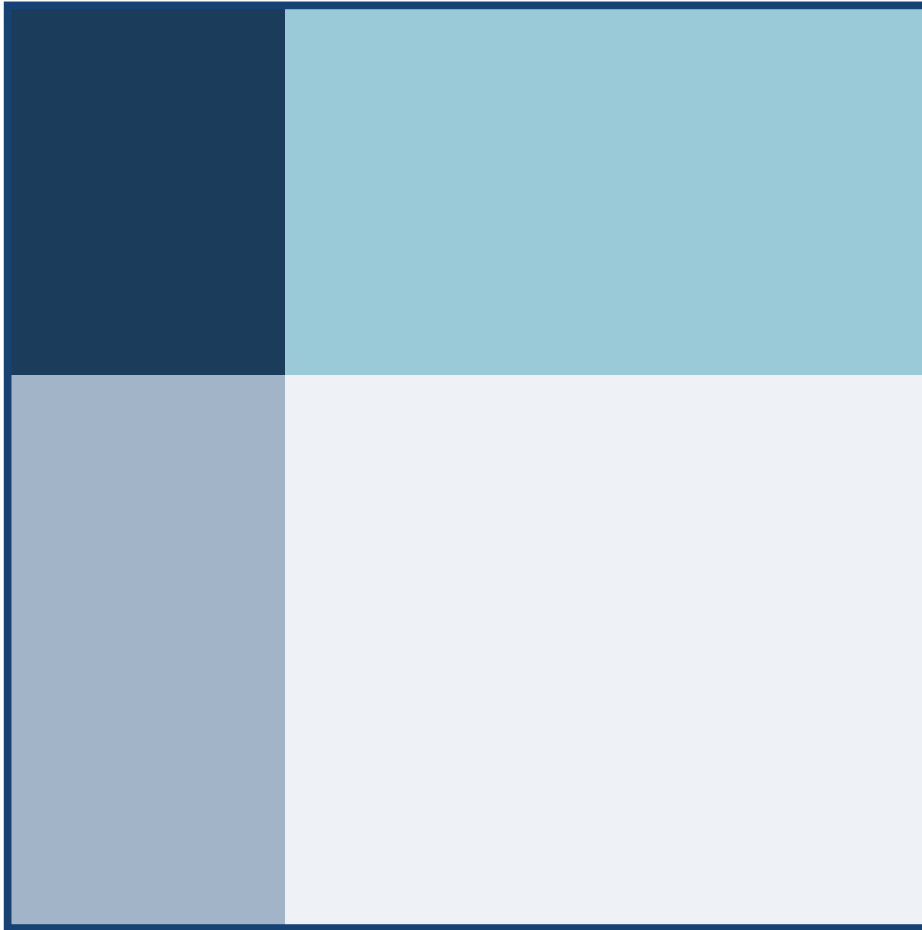
Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

Independence Is About Areas, Not Overlap

$$0.3 \cdot 0.4 = 0.12 = P(A \cap B) \rightarrow \text{independent}$$

Ω



$$P(A) = 0.3$$

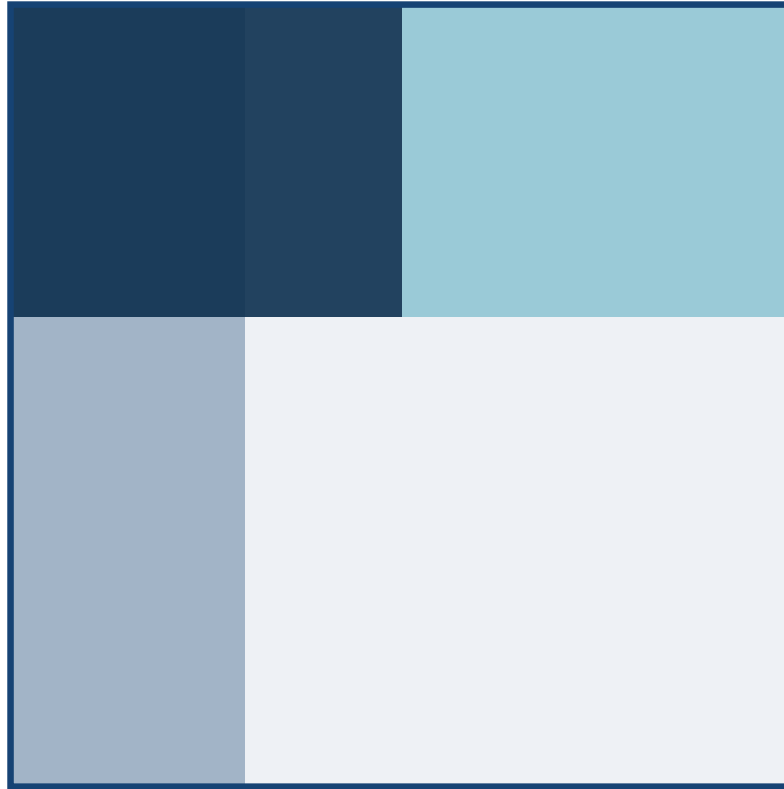
$$P(B) = 0.4$$

$$P(A \cap B) = 0.12$$

The Same Picture, Dependent

$$0.3 \cdot 0.4 = 0.12 \neq 0.2 = P(A \cap C) \rightarrow \text{dependent}$$

Ω



$$P(A) = 0.3$$

$$P(C) = 0.4$$

$$P(A \cap C) = 0.2$$

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

More Than Two Events

Events $A_1, \dots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \dots, i_l$:

$$P(A_{i_1} \cap \dots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Takeaway

Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Conditional Independence

Definition. Events A and B are **conditionally independent given C** (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A .**

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$$\frac{1}{16} \neq \frac{3}{32}, \text{ so } A \not\perp B$$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

$$\text{so } A \perp B \mid C$$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

Why This Matters

Explaining away

A common cause C can make two effects look related.

Model size

Conditional independence lets a joint distribution factorise into small pieces.

Later in the course

Naive Bayes assumes it.
Graphical models are built from it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Takeaway

Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

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Your test comes back positive. How likely is it that you are actually sick?

Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

$$P(\text{sick} \mid \text{positive}) = 1/11 \approx 9\%$$



What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

Diagram illustrating the components of Bayes' Theorem:

- posterior** — what we believe after (points to $P(A | B)$)
- likelihood** (points to $P(B | A)$)
- evidence** (points to $P(B)$)
- prior** — belief before the evidence (points to $P(A)$)
- new data** (points to $P(B)$)
- explains B** (points to $P(B | A)$)

Which of the four was "99% accurate", and which one did the intuition ignore?

Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Bayes' Theorem — Multiple Hypotheses

Theorem (Bayes, multiple events). Let $A_i \in \mathcal{P}(\Omega)$ be mutually exclusive with $\Omega = A_1 \cup \dots \cup A_k$ and $P(A_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(A_i | B) = \frac{P(B | A_i) \cdot P(A_i)}{P(B)} = \frac{P(B | A_i) \cdot P(A_i)}{\sum_{j=1}^k P(B | A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^k P(B \mid A_j) \cdot P(A_j)$$

Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate $P(\text{sick} \mid \text{positive})$ before computing it.

Mini-Exercise — Solution

Mini-Exercise — Solution

Out of 1,000: 50 sick $\rightarrow \approx 50$ true positives 950 healthy $\rightarrow \approx 10$ false alarms.

$$P(\text{sick} \mid \text{positive}) \approx \frac{50}{60} \approx 83$$

The test did not change. The prior did. This is why screening decisions depend on who is screened.

Takeaway

Bayes' theorem is the arithmetic of changing your mind: **posterior** $\propto$ **likelihood** $\times$ **prior**.
Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Module 5

Random Variables and Distributions

Guiding Question: *We rarely care which outcome occurred, only what it was worth. How do we move from outcomes to quantities?*

Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins: $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$, eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads** Y :

$P(Y = 0) = \frac{1}{8}$	$P(Y = 1) = \frac{3}{8}$	$P(Y = 2) = \frac{3}{8}$	$P(Y = 3) = \frac{1}{8}$
(T, T, T)	$(H, T, T), (T, H, T), (T, T, H)$	$(H, H, T), (T, H, H), (H, T, H)$	(H, H, H)

Y compresses eight outcomes into four numbers — and the probabilities come along automatically.

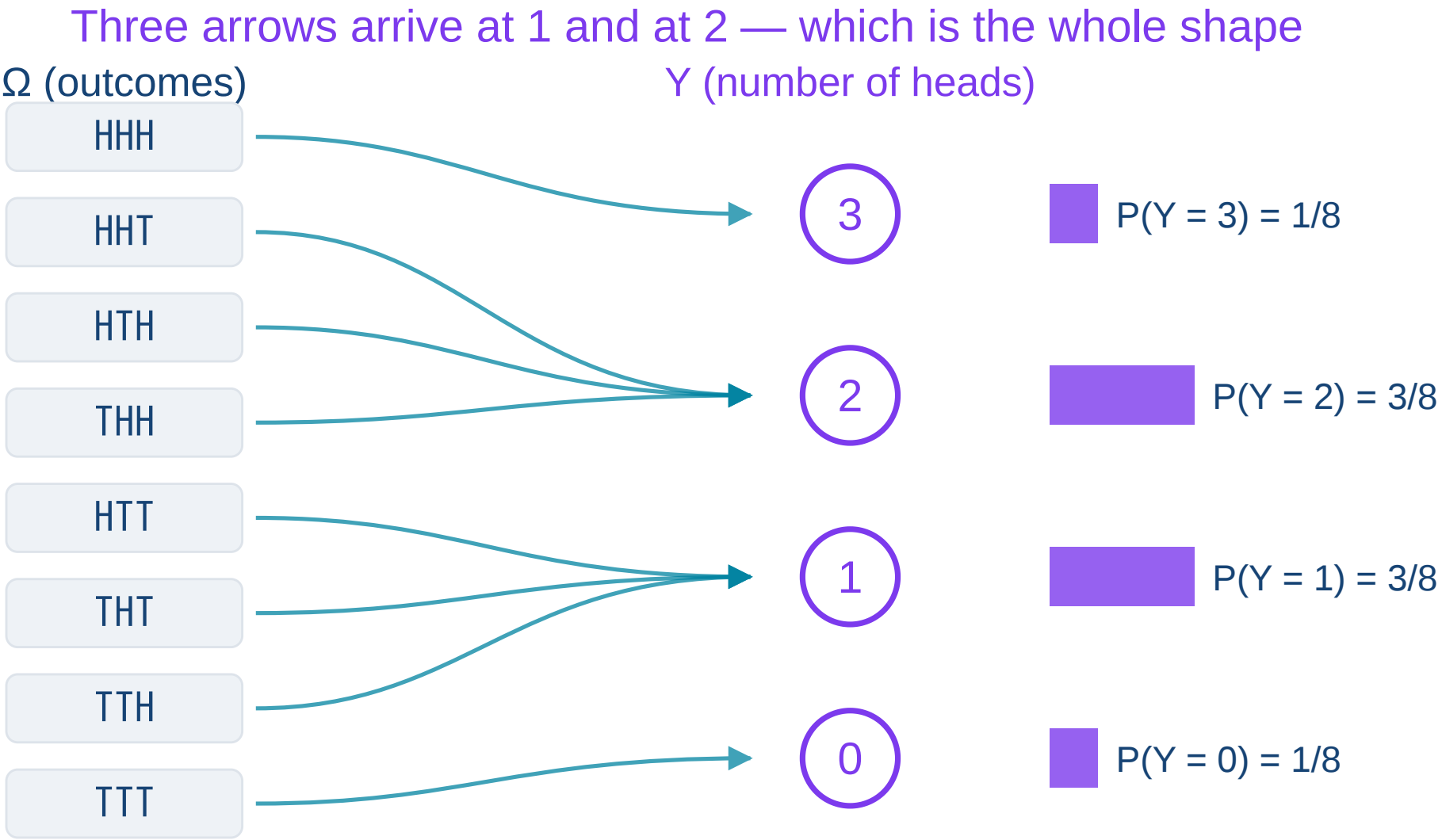
Definition

Definition (Random variable). A random variable X maps outcomes to a measurable range E : $X : \Omega \rightarrow E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω , not in X .

The Function, Drawn



Examples

Two coins

$$\Omega = \{HH, HT, TH, TT\}$$

X = number of heads

People

$$\Omega = \{\text{person}_1, \text{person}_2, \dots\}$$

X = height of the person

Two dice

$$\Omega = \{\langle n_1, n_2 \rangle\}$$

$$X(\omega) = n_1 + n_2$$

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — the subject of the next two units.

A Random Variable With Infinitely Many Values

Flip a biased coin ($P(H) = p$) until the first head. Let X be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1 - p)p, \quad P(X = 3) = (1 - p)^2 p$$

$$P(X = n) = (1 - p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X , the **probability mass function** is

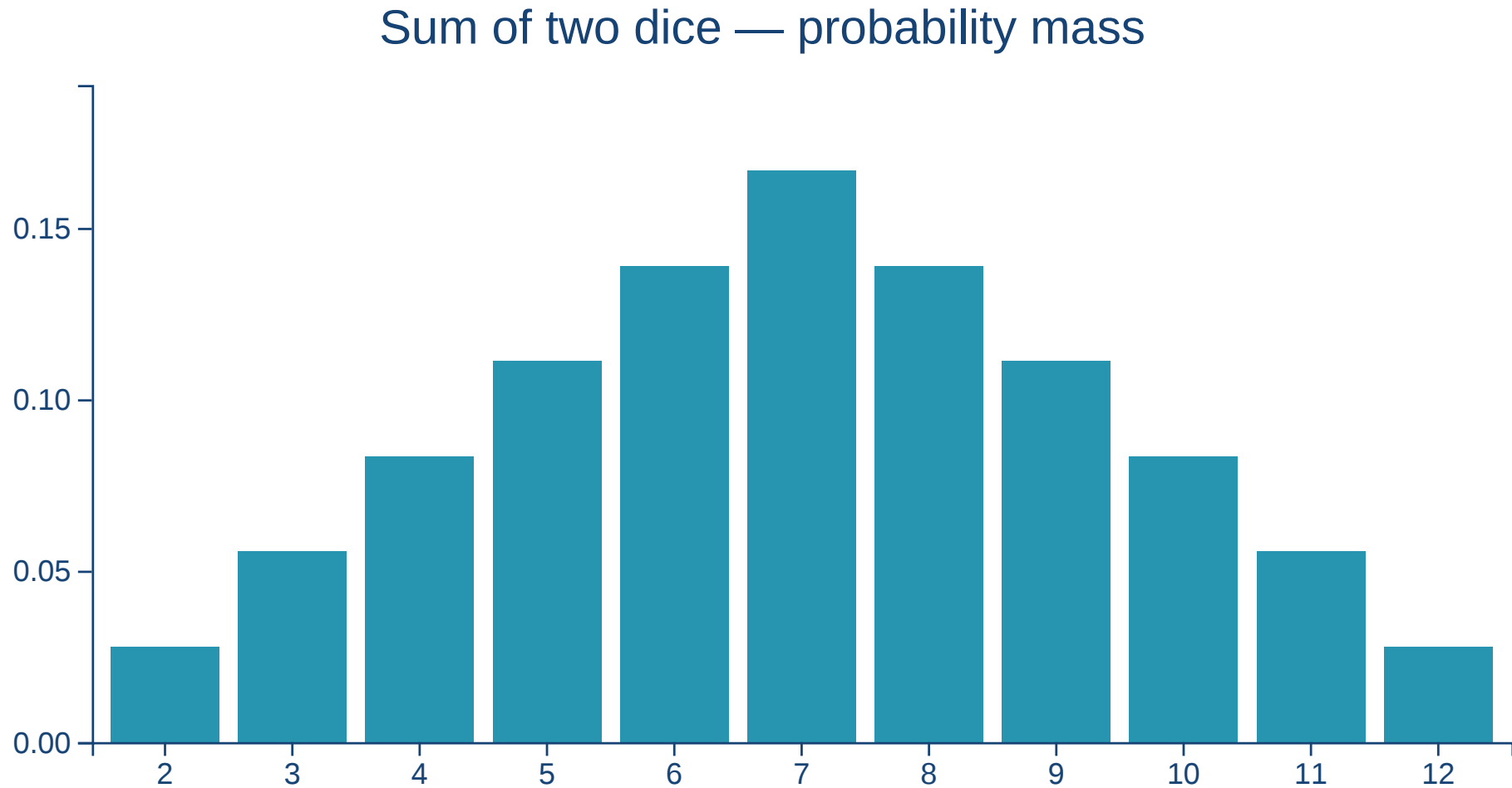
$$p(a) = P(X = a)$$

$p(a) > 0$ for at most countably
many a

$$\sum_i p(a_i) = 1$$

p is itself a probability measure

Example: Sum of Two Dice



The triangular shape is the counting from earlier units, drawn: six of the 36 outcomes give 7, only one gives 2.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment: $\mu = E[X^1]$ the first *central* moment $E[X - E[X]]$ is always 0

For a real function g :

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants a, b :

$$E[aX + b] = aE[X] + b$$

Variance

Definition (Variance).

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\sigma = \sqrt{\text{Var}(X)}$ is the **standard deviation**, in the same unit as X

Bernoulli and Binomial

Bernoulli(p) — one trial

$X = 1$ success, $X = 0$ failure

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$

Binomial $B(n, p)$ — n independent trials

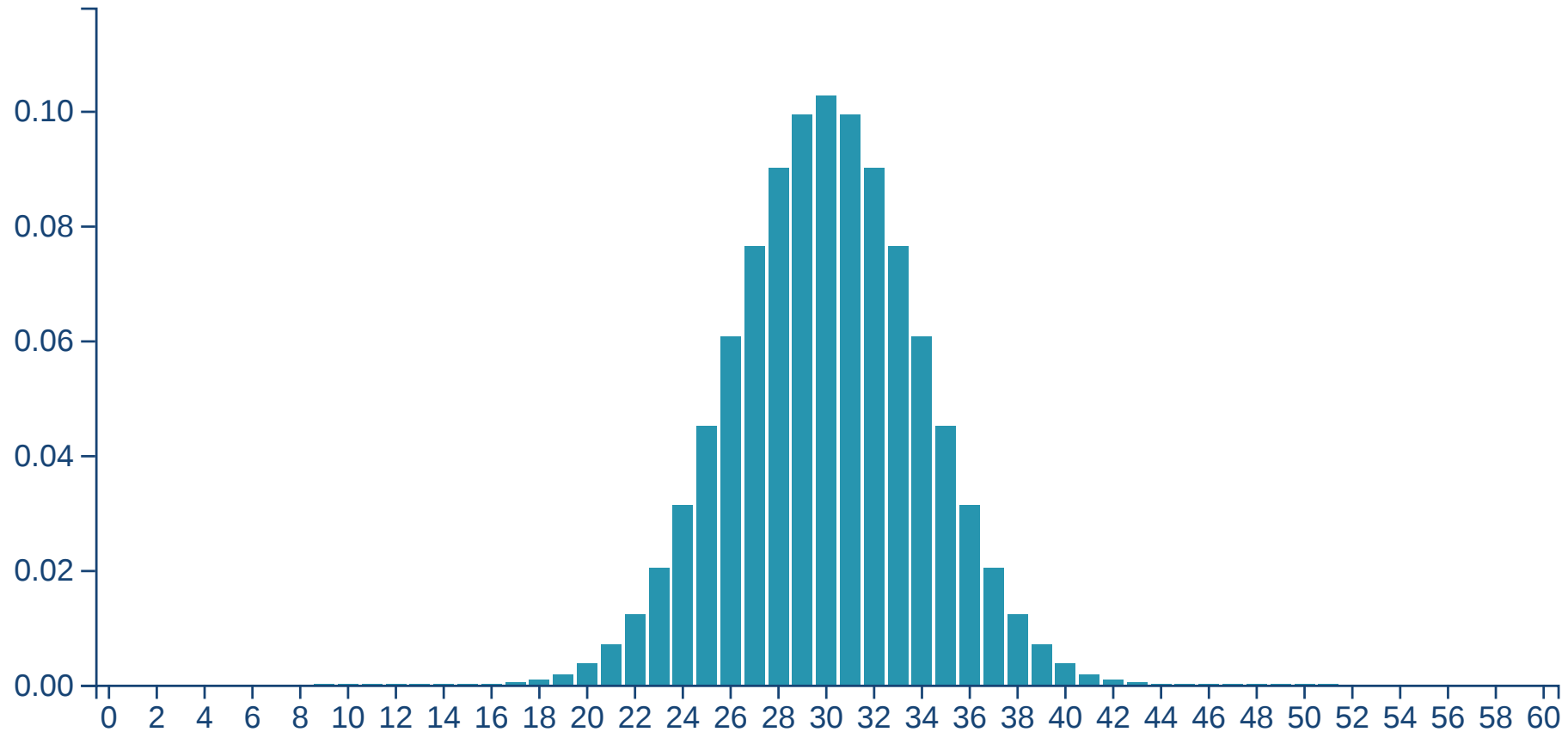
$$p(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

The binomial PMF is the counting formula times the probability of one specific sequence — combinatorics and independence in a single line.

How the Shape Depends on the Parameters

Binomial $B(n = 60, p = 0.5)$ — probability mass



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Mini-Exercise

A quiz has 10 questions with 4 options each. A student guesses every answer.

1. Which distribution describes the number of correct answers?
2. What is the expected number correct?
3. Is passing (≥ 5 correct) likely?

Mini-Exercise — Solution

Mini-Exercise — Solution

1. $B(10, 0.25)$ — ten independent trials, constant success probability
2. $E[X] = 10 \cdot 0.25 = 2.5$
3. $P(X \geq 5) \approx 0.078$ — under 8%

The independence assumption is what makes this a binomial ask when it would fail in a real exam.

Takeaway

A discrete distribution is a PMF expectation and variance compress it into **where it sits** and **how much it spreads**. Bernoulli and binomial are the counting units in probabilistic clothing.

From Mass to Density

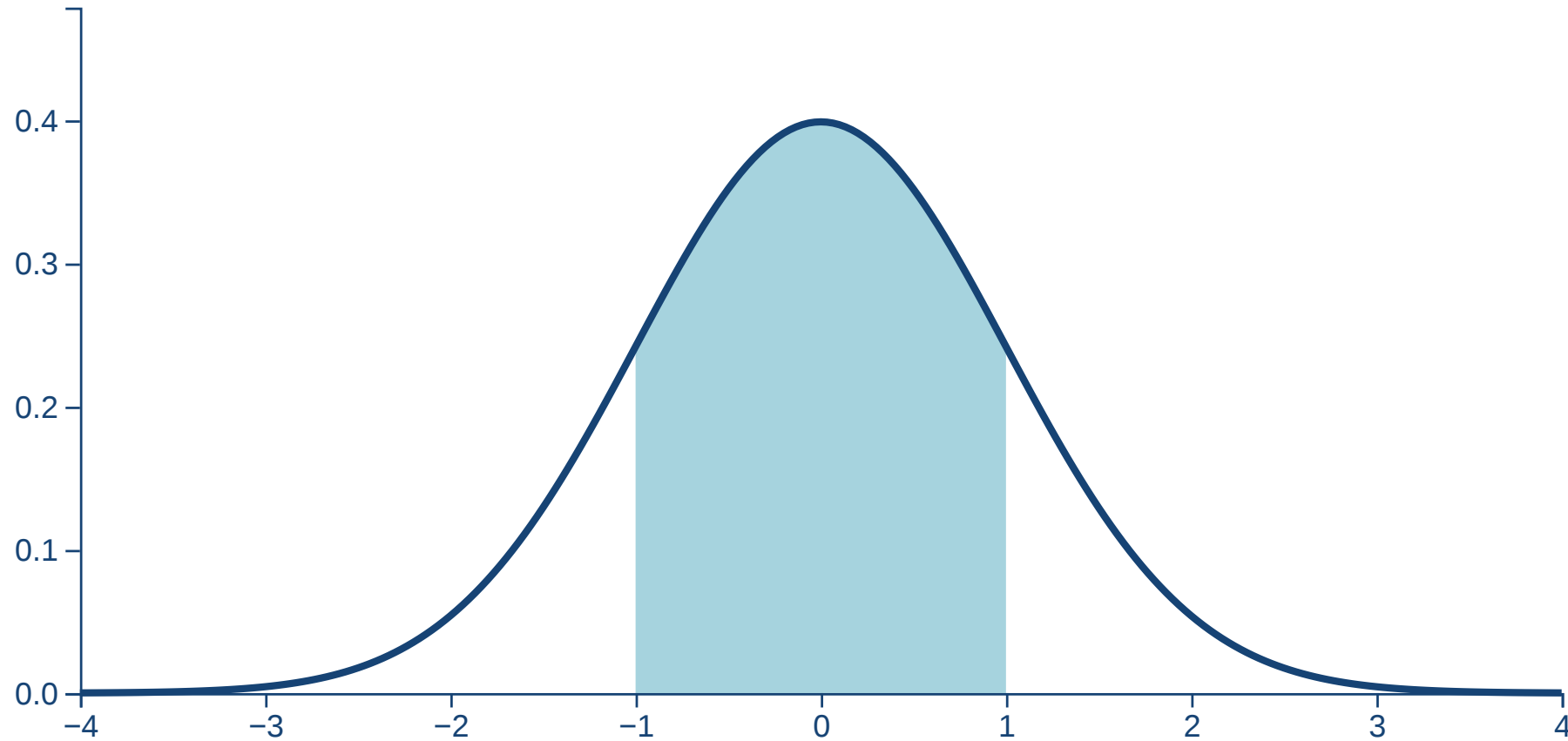
Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \leq X \leq c) = \int_b^c f(a)da, \quad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a .

Probability Is Area Under the Curve

Normal $N(\mu = 0, \sigma = 1)$ — density



The shaded region is $P(-1 \leq X \leq 1)$ — the same "probability as area" intuition from the axioms of probability, now with a curve instead of a Venn diagram.

Expectation and Variance — Sums Become Integrals

Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged: $E[aX + b] = aE[X] + b$, and $\text{Var}(X) = E[(X - \mu)^2]$.

Two Basic Continuous Distributions

Uniform on $[a, b]$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

Every interval of equal length is equally likely — the Laplace assumption, continuous version.

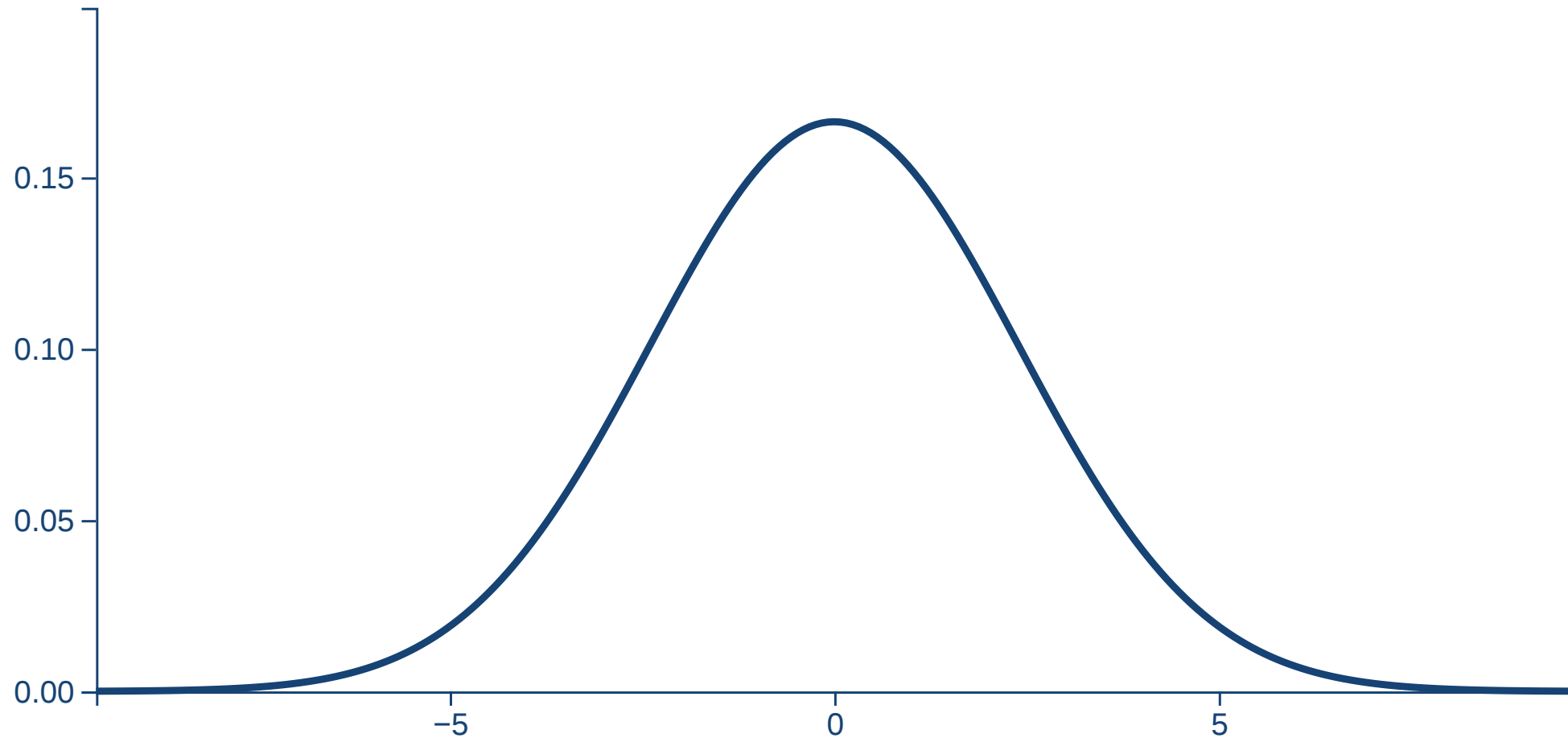
Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$, $\text{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

What σ Does to the Shape

Normal $N(\mu = 0, \sigma = 2.4)$ — density



The peak falls as the curve widens — the area is always 1.

Why the Normal Is Everywhere

The **central limit theorem**: under mild conditions, averages of many independent observations converge to a normal distribution — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown, and why the binomial started looking bell-shaped as n grew in the previous unit.

Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until n events have occurred
- **Beta** — values confined to a finite interval $[a, b]$
- **Dirichlet** — distributions over probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

More Than One Random Variable

Univariate

one random variable

Bivariate

two random variables

Multivariate

three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Joint Mass and Density

Discrete — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

Continuous — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b) da db$$

Notational convention from here on. We stop distinguishing discrete from continuous variables and simply write $p(a,b,c,\dots)$ for the probability that $\{X_1, X_2, X_3, \dots\}$ is in state $a, b, c, \dots$

Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \dots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \dots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability $\frac{1}{2}$.

Let B = number of boys, G = number of girls.

$p(B, G)$	$G = 0$	$G = 1$	$G = 2$	$G = 3$	$p(B)$
$B = 0$	.15	.10	.0875	.0375	.375
$B = 1$	.10	.175	.1125	0	.3875
$B = 2$	.0875	.1125	0	0	.2
$B = 3$	.0375	0	0	0	.0375
$p(G)$	.375	.3875	.2	.0375	1

The margins of the table *are* the marginals — which is where the name comes from.

Covariance

$$\text{cov}(A, B) = E[(A - E[A])(B - E[B])]$$

$$\text{cov} > 0$$

A large when B large —
positively related

$$\text{cov} < 0$$

A large when B small —
negatively related

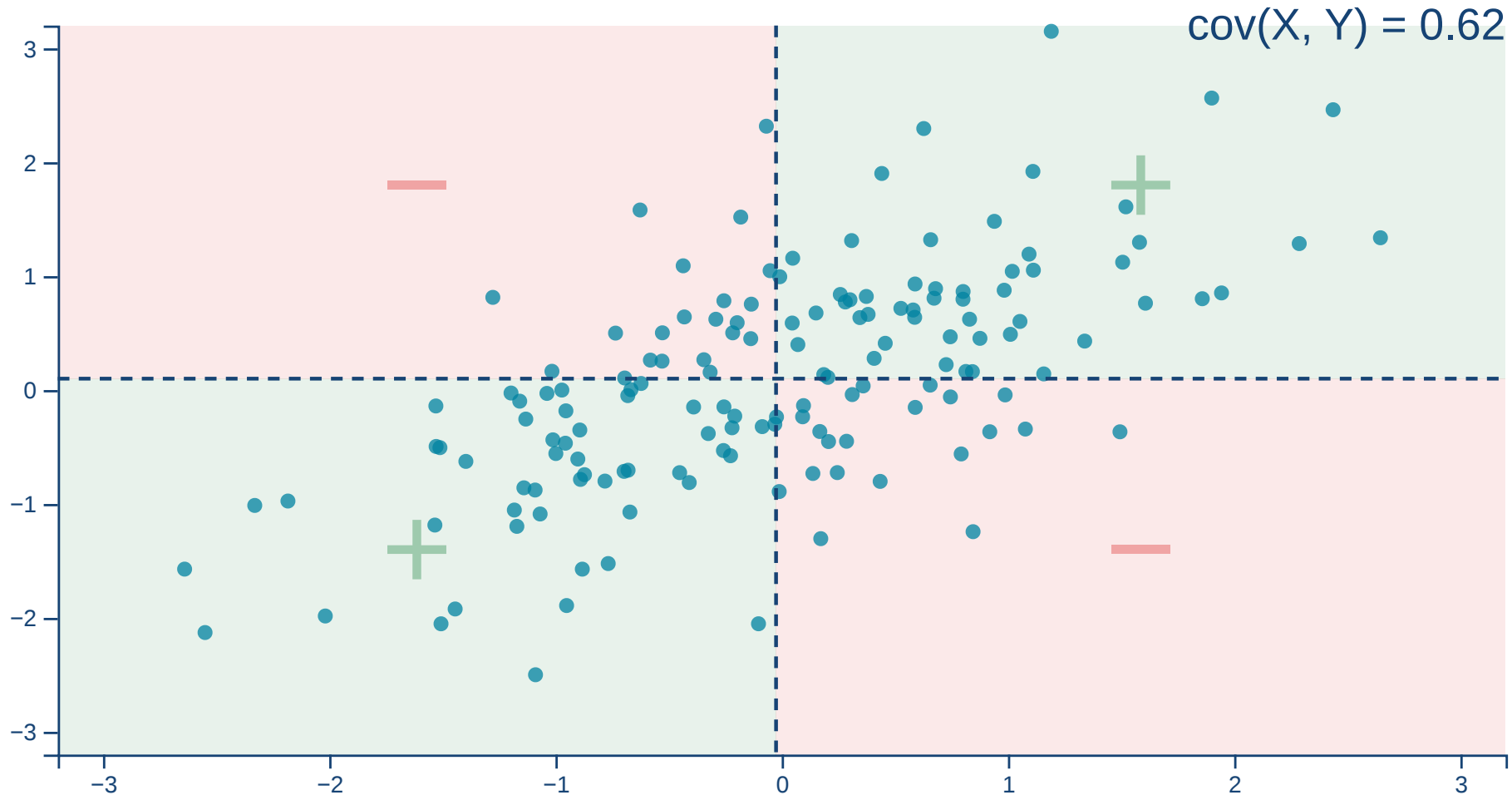
$$\text{cov}(A, A) = \text{Var}(A)$$

variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

Covariance Sees Only One Kind of Structure

Slide ρ , then switch the relation. Predict the covariance first.



Predict first: with $Y = X^2 - 1$, every point is fixed by its x . The covariance?

Takeaway

The joint distribution holds everything **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 6

Describing Data

Guiding Question: *The distribution is unknown and all we have is a sample. What can we honestly say about it?*

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

Population

The entire group — all humans on earth, all transactions, all documents.

Its properties (μ, σ^2) are the truth we cannot observe.

Sample

The observations we actually have.

Its properties $(\bar{x}, s^2)$ are what we compute — estimates of the population's.

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

What We Summarise

Central tendency

1st order moment
sample mean, median

Dispersion

2nd central moment
variance, standard deviation, IQR

Shape

3rd central moment
skewness, quantiles

Univariate statistics describe one random variable **multivariate** statistics describe $X_1, \dots, X_d$ together.

Means

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Weighted mean

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Geometric mean — for rates and ratios

$$\left(\prod_i x_i \right)^{1/n}$$

Harmonic mean — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

Trimmed mean: drop the k smallest and largest values, then average — a cheap way to buy robustness.

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$ -th value, for even n the average of the two middle values.

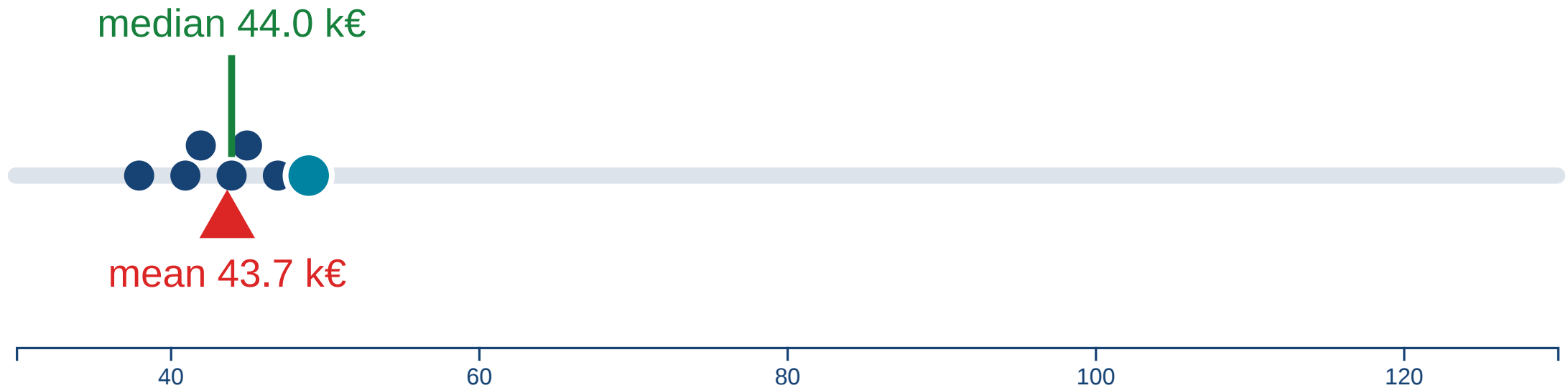
Robust against outliers — one extreme value moves the median by at most one position

Costlier to compute: requires sorting, or an interval approximation over binned data

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?



Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Mini-Exercise — Solution

Mean ≈ 73.9 , median = 44.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Dispersion describes how stretched or squeezed the data is around its centre.

Sample variance — the variance *of the observed data*

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Standard deviation $s = \sqrt{s^2}$, in the unit of the data.

Computable in **one pass** over the data, $O(n)$, using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. If the true mean μ is unknown, we must estimate it — and using $\bar{x}$ in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Dividing by $n - 1$ compensates: one degree of freedom was spent estimating the mean from the same data.

Takeaway

s^2 describes **the sample** $\hat{\sigma}^2$ estimates **the population**. The $n - 1$ is not a convention — it is the price of having estimated the mean from the same data.

Quantiles

The k -th q -**quantile** is the value x such that k/q of the values are smaller and $(q - k)/q$ are larger ($0 < k < q$).

Percentiles

$$q = 100$$

"the 12th percentile"

Quartiles

$$q = 4$$

$Q_1 = 25\text{th}$, $Q_2 = 50\text{th}$ (= median),

$Q_3 = 75\text{th}$ percentile

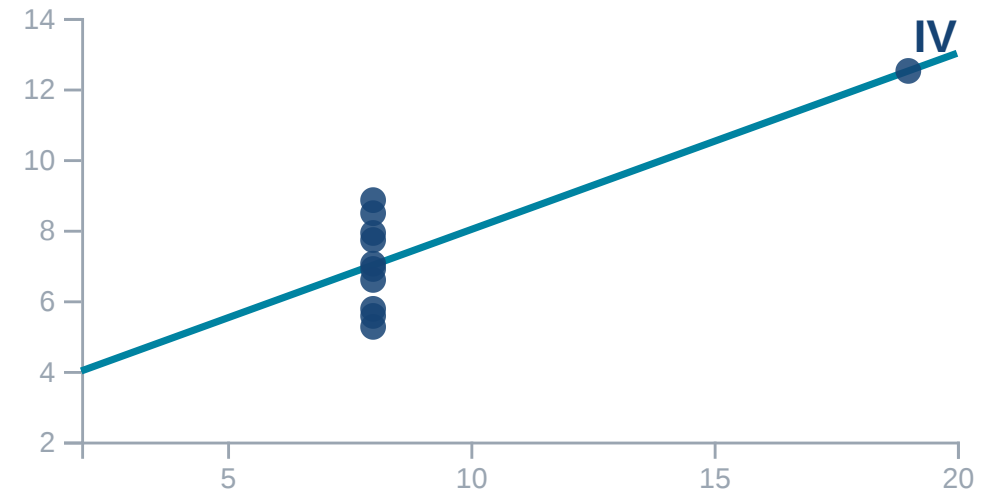
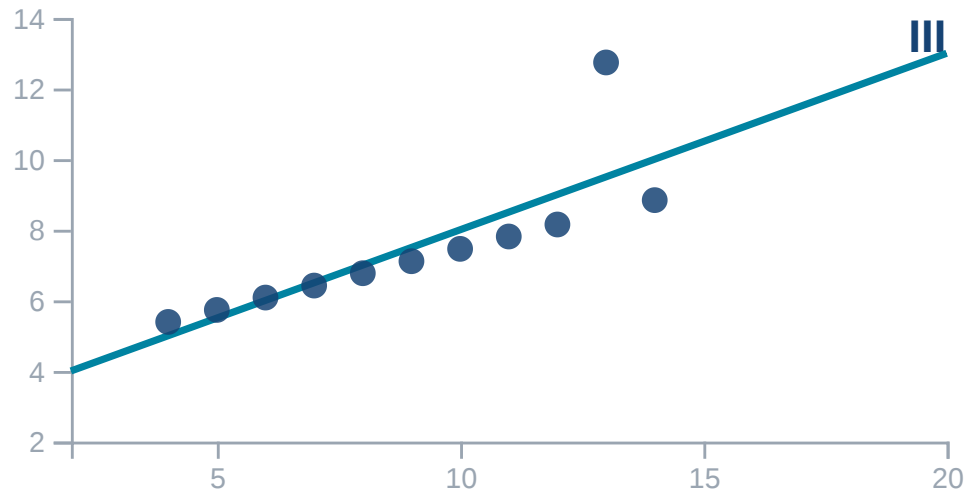
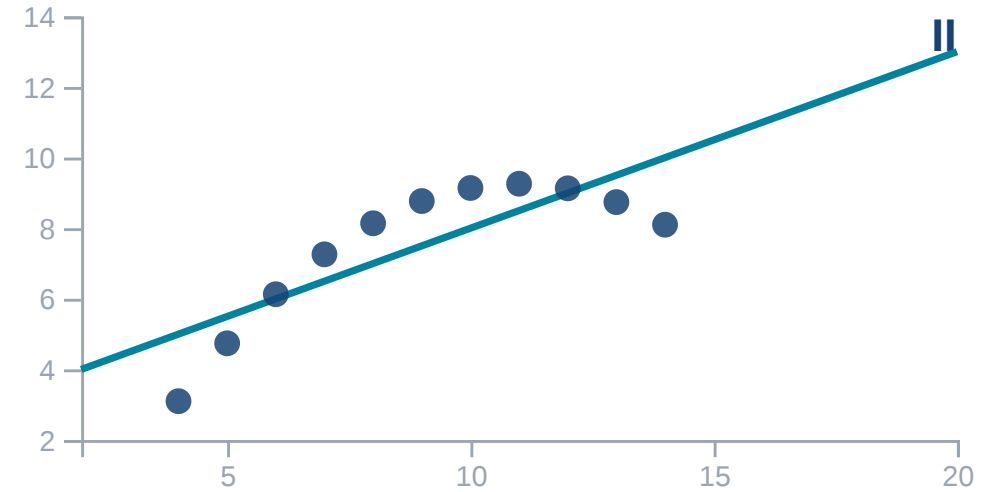
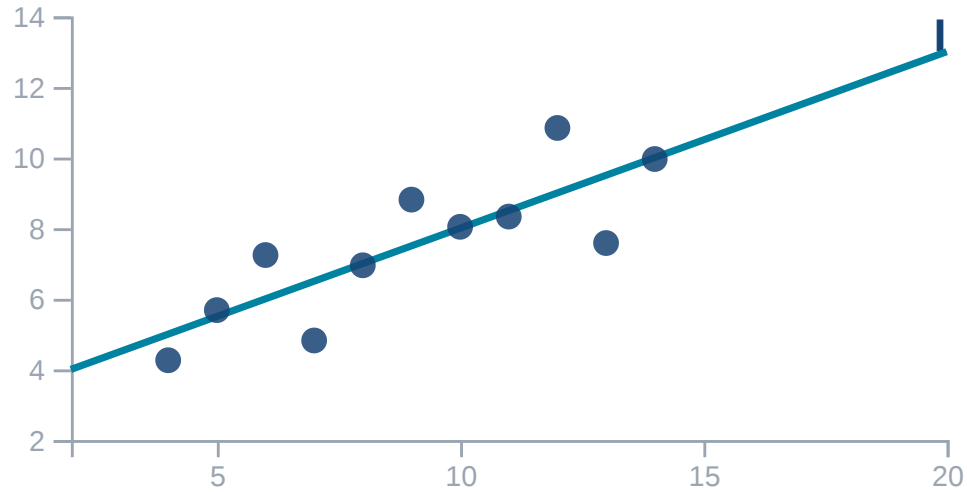
IQR

$$\text{IQR} = Q_3 - Q_1$$

the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data



What the Quartet Teaches

Summaries compress — and compression loses.

Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.

Visual inspection is not a soft alternative to statistics it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

Module 7

Hypothesis Testing

Guiding Question: *Your new model scores higher than the baseline. How do you know the improvement is real?*

Motivation: Comparing Two ML Experiments

Experiment A — baseline model
accuracy measured over n_A runs

Experiment B — new architecture
accuracy measured over n_B runs

B scored higher. Is the improvement systematic, or just randomness?

H_0 : the two means are equal — $\mu_A = \mu_B$

H_1 : they differ — $\mu_A \neq \mu_B$

The Idea

Compare what we **observed** with what H_0 **predicts**. If the disagreement is large enough to be implausible under H_0 , reject H_0 .

We never prove H_1 . We only measure how badly H_0 fails to explain the data.

Test Statistic and p-Value

A **test statistic** $T = T(X_1, \dots, X_n)$ compresses the data so that its distribution **under** H_0 is known.

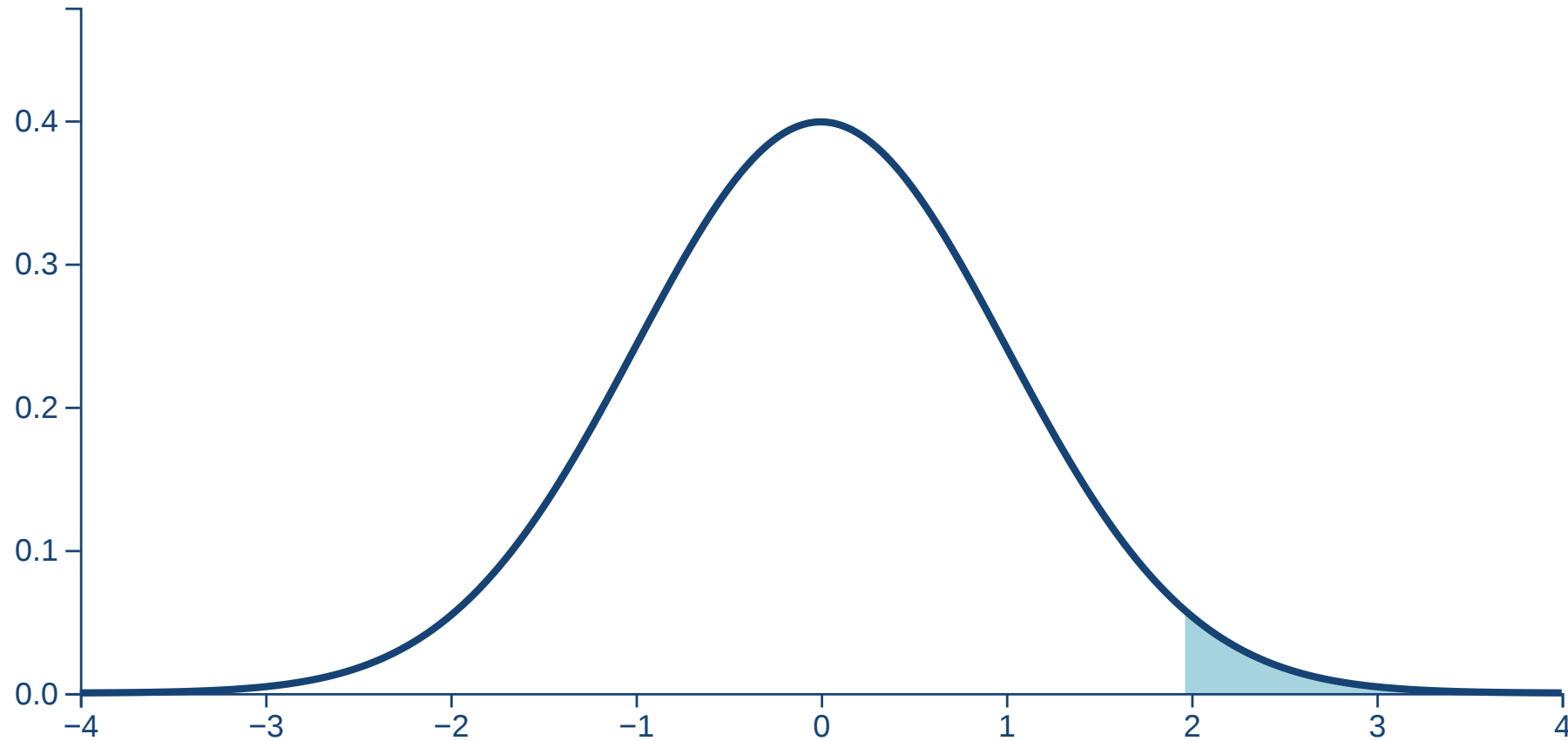
The **p-value** is the probability, under H_0 , of observing a statistic at least as extreme as the one we got.

Small p-value $\rightarrow$ the data is surprising under H_0 .

The p-value is **not** the probability that H_0 is true. It is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$.

The Rejection Region

Normal $N(\mu = 0, \sigma = 1)$ — density



Under H_0 the statistic follows a known distribution. The shaded tail carries probability $\alpha/2$: if T lands there, we reject.

Two Ways to Be Wrong

Type I error — false alarm

Reject H_0 when it is true.

Probability α (we choose it).

Type II error — miss

Fail to reject H_0 when H_1 holds.

Probability β **power** = $1 - \beta$.

Lowering α raises β for a fixed sample size. More data is the only way to improve both.

Which Test?

One sample, known variance — z -test

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Reject (two-sided) if $|Z| > z_{1-\alpha/2}$.

One sample, unknown variance — t -test

$$T = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

Same standardised distance, estimated denominator → **heavier tails**.

Two samples, equal variances — pooled t

Use when variance homogeneity is credible (same pipeline, same noise).

Two samples, unequal variances — Welch's t

The default for ML A/B: different randomness induces different variances.

Testing Variances

Single variance (normal model), χ^2 statistic with $n - 1$ degrees of freedom.

Is run-to-run variability within specification?

Two variances, $F = S_A^2 / S_B^2$.

If the variances differ, **do not pool** — go back to Welch's t for the means.

Back to the A/B Question

Default

Welch's two-sample t on the mean difference

Care about stability

F -test on S_A^2 / S_B^2

Normality dubious

Permutation (randomisation) test on the mean difference

Hypothesis testing compares data to a model through a statistic whose null distribution is known. Known vs. unknown variance changes that distribution (z vs. t) unequal variances change the test (Welch, not pooled).

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.